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Nuclear fission plants

Assignment 3:

Thermo-mechanical verification of a fuel pin of an AP1000 reactor

Master degree in Sustainable nuclear energy

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Chapter 1

Introduction

This assignment is focused on the thermo-mechanical stresses that fuel rods experience inside a nuclear reactor. The aim of this report is to evaluate these stresses due to different loads (pressure, thermal, bending) and to perform a verification of them according to the ASME BPVC Section III standard. The main load considered is the pressure, because it is the most dangerous one, while the thermal load gives a smaller contribution and bending can be neglected. In the end, it will be possible to size the gas plenum to reduce the internal pressure down to acceptable values. The reactor design in this assignment is the AP1000 and the operating conditions are the same as in Assignment 2.

1.1 Assumptions and input data

The analysis is performed considering some assumptions:

- Reactor operating at steady-state in nominal conditions;
- Only static loads are considered;
- Bending stresses, peak stresses and local stresses are neglected;
- Cladding material is Zircaloy-4;
- The maximum internal pressure p_i is estimated through Mariotte's formula in the conservative hypothesis of a depressurized reactor ($p_o \sim 0$). This bounding p_i is then used to evaluate the mechanical stress field, which is computed under nominal operating conditions with the external coolant pressure set to $p_o = p_{\text{sys}}$.

As previously mentioned, the input data are taken from the analysis in Assignment 2. Since in this case the cladding material is Zircaloy-4 instead of ZIRLO, the material properties are taken from [1].

Table 1.1: Input parameters used in the cladding structural analysis [2].

Symbol	Description	Value	Unit	SI value	SI unit
<i>Operating conditions</i>					
p_{sys}	System pressure	2250	psia	1.551×10^7	Pa
<i>Geometry (square 17×17 array)</i>					
D_{co}	Outer cladding diameter	0.374	in	9.500×10^{-3}	m
s_{clad}	Cladding thickness	0.0225	in	5.715×10^{-4}	m
D_f	Fuel pellet diameter	0.3225	in	8.192×10^{-3}	m
H_{active}	Active core height	168	in	4.267	m
<i>Material properties (Zircaloy-4)^a</i>					
σ_y	Yield stress	241	MPa	2.410×10^8	Pa
σ_u	Ultimate stress	413	MPa	4.130×10^8	Pa

^a Properties from [1], Tab. 2.

^b $D_{ci} = D_{co} - 2 s_{\text{clad}}$.

Table 1.2: Temperature-dependent mechanical properties of Zircaloy-4 [1].

Property	Correlation (Zircaloy-4)
Thermal expansion coefficient α [K^{-1}]	$\frac{6.72 \times 10^{-6} T - 2.07 \times 10^{-3}}{T - 308}$
Young's modulus E [MPa]	$[9.9 \times 10^3 - 5.669 (T - 273)] 9.81$
Poisson's ratio ν [—]	$0.3303 + 8.376 \times 10^{-5} (T - 273)$

* All temperatures are expressed in K.

Chapter 2

ASME verification

This chapter is dedicated to the analysis of thermo-mechanical stresses of fuel pin cladding during the operations. The main loads considered are pressure, thermal and bending. The first load, pressure, is the most dangerous and is here firstly accounted with the buckling. Then the analysis is completed with the evaluation of mechanical and thermal stresses. The last step is a verification according to the ASME BPVC Section III of the values retrieved.

2.1 Buckling analysis

The first phenomenon that has to be verified is the buckling of the rods, which causes deformation and geometrical instability. The main cause is a compression due to the pressures acting on the rod. In order to avoid buckling, it is necessary to verify that the applied pressure remains below the critical buckling pressure p_{cr} :

$$p_{cr} = \frac{E}{4(1 - \nu^2)} \left(\frac{t}{r_{avg}} \right)^3 \quad (2.1)$$

where t is the cladding thickness and r_{avg} is the average distance of the cladding from the fuel pellet centerline. The Young's modulus E and the Poisson's ratio ν are evaluated through the Zircaloy-4 correlations reported in Table 1.2.

As shown in Figure 2.1, the critical buckling pressure remains in the range 45.97–48.88 MPa along the active length, well above the system pressure ($p_{sys} = 15.51$ MPa), so the cladding is verified against buckling.

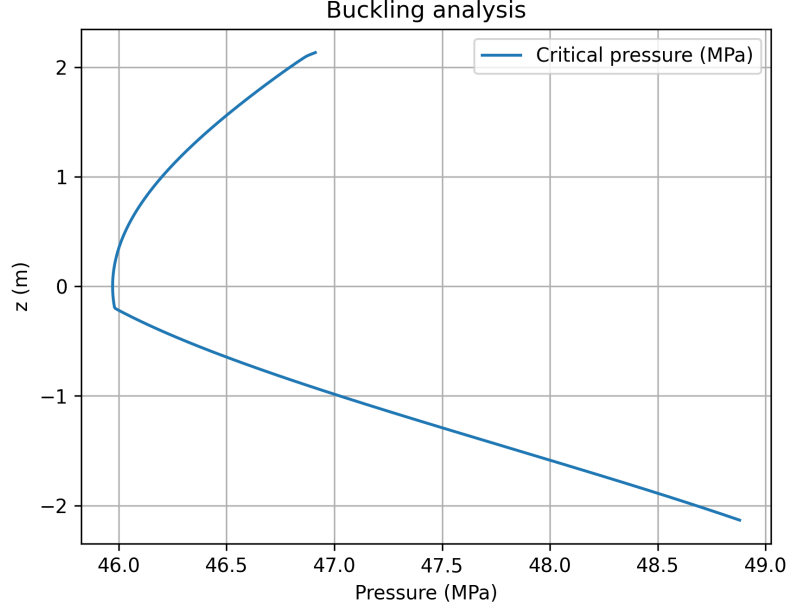


Figure 2.1: Critical buckling pressure p_{cr} along the axial coordinate z .

2.2 Internal pressure

The calculation of the internal pressure is necessary to compute the mechanical and thermal stresses. To evaluate the maximum internal pressure we assume, only in this case, a depressurized reactor. For this estimate the cladding internal pressure is taken at its maximum value and is evaluated with the Mariotte formula:

$$p_i = \frac{\sigma_h t}{r_{avg}} \quad (2.2)$$

Mariotte is applicable because the thin-tube assumption ($\frac{t}{r} \ll 1$) is valid. σ_h is the tangential stress, which in this bounding case is taken equal to the yield stress σ_y given as input. This yields a maximum internal pressure $p_i = 30.85$ MPa, which is used to evaluate the mechanical stress field of the following section.

2.2.1 Mechanical Stress Analysis

The mechanical stresses acting on the cladding are classified as primary stresses (P) according to the ASME BPVC Section III standard, as they are directly induced by external mechanical loads and are necessary to satisfy the equilibrium of forces. The cladding is modeled as a thick-walled hollow cylinder subject to an internal fission gas pressure p_i and an external coolant operating pressure $p_o = p_{sys}$.

The primary stress field consists of tangential (hoop) σ_h , radial σ_r , and longitudinal (axial) σ_l components. According to Lamé's equations for thick cylinders, these components vary radially and are evaluated at the inner wall ($r = r_{in}$) and outer wall ($r = r_{out}$). At the inner wall:

$$\sigma_{h,in} = \frac{p_i(r_{in}^2 + r_{out}^2) - 2p_o r_{out}^2}{r_{out}^2 - r_{in}^2} \quad (2.3)$$

$$\sigma_{r,in} = -p_i \quad (2.4)$$

At the outer wall:

$$\sigma_{h,out} = \frac{2p_i r_{in}^2 - p_o(r_{in}^2 + r_{out}^2)}{r_{out}^2 - r_{in}^2} \quad (2.5)$$

$$\sigma_{r,out} = -p_o \quad (2.6)$$

The longitudinal component is uniform across the cladding thickness, $\sigma_{l,in} = \sigma_{l,out} = \sigma_l$:

$$\sigma_l = \frac{p_i r_{in}^2 - p_o r_{out}^2}{r_{out}^2 - r_{in}^2} \quad (2.7)$$

The results for both inner and outer wall are shown in Table 2.2.

ASME Code Classification and Primary Stresses

Per the ASME code framework, the general primary membrane stresses (P_m) must balance the mechanical loads across the entire section, excluding localized discontinuity effects. To calculate P_m , the average values of the stress components across the cladding thickness are computed:

$$\sigma_{avg,j} = \frac{\sigma_{j,in} + \sigma_{j,out}}{2} \quad \text{for } j \in \{h, r, l\} \quad (2.8)$$

The secondary mechanical stress contributions, representing the deviation from the mean membrane state, are subsequently evaluated to capture local self-equilibrating states:

$$\sigma_{j,sec} = \sigma_j - \sigma_{avg,j} \quad (2.9)$$

Applying Tresca's maximum shear stress theory, the equivalent primary stress intensity P_{avg} is evaluated from the average membrane tensor:

$$P_{avg} = \max(|\sigma_{avg,h} - \sigma_{avg,r}|, |\sigma_{avg,r} - \sigma_{avg,l}|, |\sigma_{avg,l} - \sigma_{avg,h}|) \quad (2.10)$$

Under design conditions, the compliance check dictates that $P_{avg} \leq S_m$, where the allowable stress intensity limit S_m is determined by the material properties at temperature:

$$S_m = \min\left(\frac{2}{3}\sigma_y, \frac{1}{3}\sigma_u\right) \quad (2.11)$$

Given $\sigma_y = 241$ MPa and $\sigma_u = 413$ MPa for Zircaloy-4, the primary design limit is established at $S_m = 137.67$ MPa.

The quantitative results of the primary membrane stress components and the final structural compliance check against the ASME BPVC Section III limits are summarized in Table 2.1.

Table 2.1: Summary of cladding average stresses and ASME verification under design conditions.

Stress Component / Parameter	Value [MPa]
Circumferential average stress ($\sigma_{avg,h}$)	97.13
Radial average stress ($\sigma_{avg,r}$)	-23.18
Longitudinal average stress ($\sigma_{avg,l}$)	36.97
Maximum principal stress intensity (σ_{max})	120.32
Allowable stress intensity limit (S_m)	137.67
Final structural verdict	✓ OK

As indicated by the numerical evaluation, the maximum principal stress intensity ($\sigma_{max} = 120.32$ MPa) satisfies the design constraint, confirming that the cladding thickness is structurally adequate against the bounding internal pressure loads.

2.2.2 Thermal Stress Analysis

Thermal stresses are classified as secondary stresses (Q) under the ASME criteria since they are self-limiting and originate from the structural constraint against differential thermal expansion across the cladding thickness. The steep radial temperature gradient $\Delta T = T_{ci} - T_{co}$ induced by the peak heat flux generates compressive stresses at the hotter inner surface and tensile stresses at the cooler outer surface.

The calculation accounts for temperature-dependent material properties. The Young's modulus $E(T)$, Poisson's ratio $\nu(T)$, and linear thermal expansion coefficient $\alpha(T)$ are evaluated at the average cladding bulk temperature $T_{avg} = (T_{ci} + T_{co})/2$ using the Zircaloy-4 correlations reported in Table 1.2.

The analytical thermo-elastic solutions for the non-vanishing thermal stress components (tangential and longitudinal) at the boundaries are given by:

$$\sigma_{h,th}^{in} = \sigma_{l,th}^{in} = \frac{E\alpha(T_{ci} - T_{co})}{2(1 - \nu) \ln\left(\frac{r_{out}}{r_{in}}\right)} \left[1 - \frac{2r_{out}^2}{r_{out}^2 - r_{in}^2} \ln\left(\frac{r_{out}}{r_{in}}\right) \right] \quad (2.12)$$

$$\sigma_{h,th}^{out} = \sigma_{l,th}^{out} = \frac{E\alpha(T_{ci} - T_{co})}{2(1 - \nu) \ln\left(\frac{r_{out}}{r_{in}}\right)} \left[1 - \frac{2r_{in}^2}{r_{out}^2 - r_{in}^2} \ln\left(\frac{r_{out}}{r_{in}}\right) \right] \quad (2.13)$$

The radial thermal stress component reduces to zero at both boundary surfaces ($\sigma_{r,th}^{in} = \sigma_{r,th}^{out} = 0$) due to the lack of radial constraints at the walls.

Primary plus Secondary Stress Intensity Verification

To perform the final safety verification ($P + Q \leq 3S_m$), the total stress field is constructed element-wise along the axial profile z by superimposing primary membrane, secondary mechanical, and secondary thermal terms:

$$\sigma_{j,tot} = \sigma_{avg,j} + \sigma_{j,sec} + \sigma_{j,th} \quad \text{for } j \in \{h, r, l\} \quad (2.14)$$

The cumulative equivalent stress intensity is calculated at both the inner and outer walls using Tresca's maximum shear criterion:

$$\sigma_{max,in}^{tot} = \max\left(|\sigma_{l,in}^{tot} - \sigma_{h,in}^{tot}|, |\sigma_{l,in}^{tot} - \sigma_{r,in}^{tot}|, |\sigma_{h,in}^{tot} - \sigma_{r,in}^{tot}|\right) \quad (2.15)$$

$$\sigma_{max,out}^{tot} = \max\left(|\sigma_{l,out}^{tot} - \sigma_{h,out}^{tot}|, |\sigma_{l,out}^{tot} - \sigma_{r,out}^{tot}|, |\sigma_{h,out}^{tot} - \sigma_{r,out}^{tot}|\right) \quad (2.16)$$

The structural integrity is guaranteed if the maximum combined stress along the entire active fuel length satisfies the limit:

$$\max_z\left(\sigma_{max,in}^{tot}(z), \sigma_{max,out}^{tot}(z)\right) \leq 3S_m \quad (2.17)$$

With $3S_m = 413.0$ MPa, this constraint ensures protection against plastic ratcheting under cyclic thermal-hydraulic operations.

The evaluation of the primary plus secondary mechanical stress profiles along the active fuel length highlights the cross-over behavior between the inner and outer cladding walls. As shown in Figure 2.2, the equivalent stress intensity derived via the Tresca maximum shear stress theory yields values ranging between approximately 105 MPa and 135 MPa.

The structural integrity of the cladding is comprehensively maintained since the maximum calculated stress intensity across the entire axial span remains far below the combined primary and secondary allowable limit of $3S_m = 413.0$ MPa. This large safety margin demonstrates that the structural thickness selected for the AP1000 fuel pin is highly conservative under nominal mechanical pressure differentials.

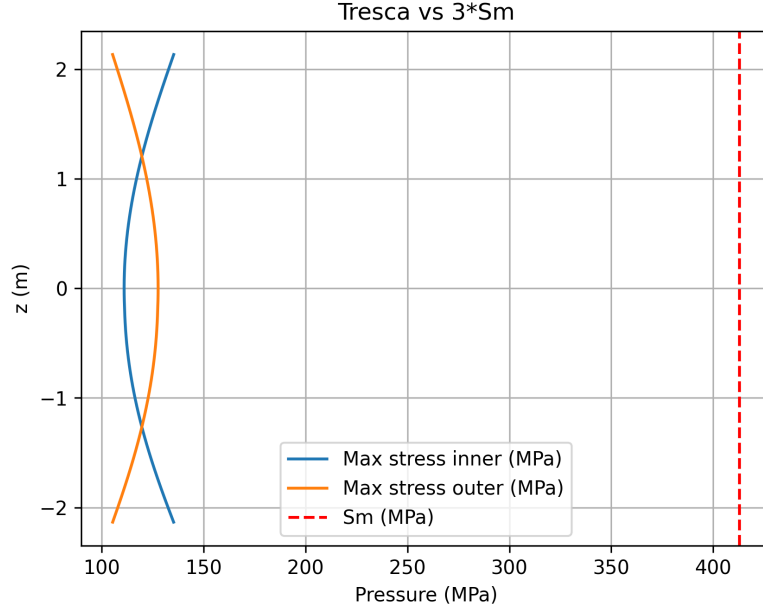


Figure 2.2: Tresca equivalent stress intensity profile along the axial coordinate z for both the inner and outer cladding walls, compared against the Service Level A limit ($3S_m = 413.0$ MPa).

The complete multi-axial stress field resulting from the element-wise superposition of mechanical and thermal loads along the core axis is illustrated in Figure 2.3, while in Table 2.2 all components are shown. The profiles showcase distinct physical behaviors for each stress component:

- **Radial Stress (σ_r):** Consistently exhibits a flat axial behavior. It perfectly matches the boundary conditions imposed by the pressures, settling at $\sigma_{r,in} = -p_i = -30.85$ MPa at the inner wall and $\sigma_{r,out} = -p_o = -15.51$ MPa at the outer wall.
- **Hoop Stress (σ_h):** The tangential component ranges between 80 MPa and 112 MPa. The steep radial thermal gradient induces compressive counter-stresses at the inner wall (shifting the curve to the left near the midplane) and tensile stresses at the outer wall (shifting the curve to the right), where the peak stress is reached due to the combined effects.
- **Longitudinal Stress (σ_l):** Shows a symmetric parabolic profile following the chopped-cosine power distribution, ranging between 12 MPa and 60 MPa. This component directly mirrors the local linear heat rate variation across the active fuel length.

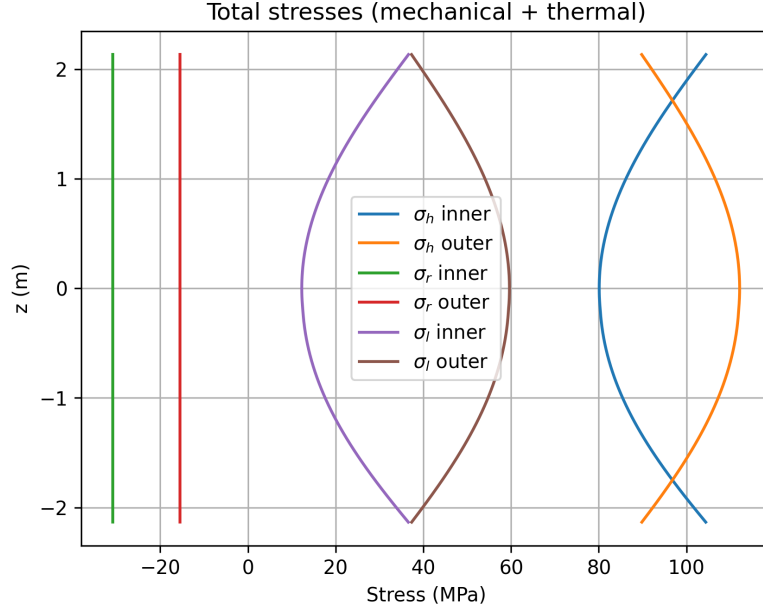


Figure 2.3: Axial distribution of total combined multi-axial stresses (tangential, radial, and longitudinal) across the inner and outer cladding boundaries under coupled thermo-mechanical loads.

Table 2.2: Mechanical, thermal and total stresses at the inner and outer cladding wall (peak-flux section, $z = 0$, $\Delta T = 59.06$ °C).

Wall	Contribution	σ_h [MPa]	σ_r [MPa]	σ_l [MPa]
Inner	Mechanical	104.80	-30.85	36.97
	Thermal	-24.69	0.00	-24.69
	Total	80.11	-30.85	12.28
Outer	Mechanical	89.46	-15.51	36.97
	Thermal	22.67	0.00	22.67
	Total	112.13	-15.51	59.65

Chapter 3

Gas plenum sizing

This chapter is dedicated to the evaluation and sizing of the gas plenum within the fuel pin, which serves to accommodate and mitigate the pressure increase resulting from the release of fission products.

Table 3.1: Input data for the gas plenum sizing.

Quantity	Symbol	Value	Unit
<i>Fuel and fission data</i>			
Burn-up ^a	BU	60,000	MW d/t
Energy per fission ^b	E_f	200	MeV
Combined fission yield (Xe+Kr)	Y	0.28	–
Release fraction	R_r	0.40	–
U-235 enrichment (weight)	e	4.45	%
UO ₂ theoretical density	ρ_{th}	10.96	g/cm ³
Fraction of theoretical density	f_{th}	0.95	–
<i>Impurities (weight concentration)</i>			
Nitrogen	c_{N_2}	25	ppm
Water vapor	c_{H_2O}	75	ppm
<i>Geometry</i>			
Pellet diameter	d_p	8.19	mm
Spring size	H_{spring}	15	cm

$$^a BU = 60000 \frac{\text{MW d}}{\text{t}_U} \cdot 10^6 \frac{\text{W}}{\text{MW}} \cdot 86400 \frac{\text{s}}{\text{d}} \cdot \frac{1}{10^3} \frac{\text{t}_U}{\text{kg}_U} = 5.184 \times 10^{12} \frac{\text{J}}{\text{kg}_U}.$$

$$^b E_f = 200 \text{ MeV} \cdot 1.60218 \times 10^{-13} \frac{\text{J}}{\text{MeV}} = 3.204 \times 10^{-11} \text{ J}.$$

Several core hypotheses and assumptions are postulated:

- Fission gases produced during reactor operation are restricted exclusively to Xenon (Xe) and Krypton (Kr).
- Impurity gases originally present in the fabricated fuel pellets consist solely of Nitrogen (N_2) and water vapor (H_2O).
- All gaseous species and vapors exhibit ideal gas behavior under operating conditions (to be verified for H_2O).
- The effective volume available for gas expansion is strictly limited to the gas plenum. The clearance gap between the fuel pellet and the cladding is neglected as available volume due to the effects of fuel pellet swelling and buckling.
- The gas plenum is assumed to be at thermal equilibrium with the cladding inner surface at that specific axial location, the temperature profile of which is retrieved from the results of Assignment 2.

Since enriched uranium fuel is considered, the uranium molar mass must be evaluated as the following weight average:

$$u_U = \frac{1}{\frac{e}{u_{235U}} + \frac{1-e}{u_{238U}}} \quad \text{and} \quad u_{UO_2} = u_U + 2 u_{16O} \quad (3.1)$$

The total volume and the total mass of uranium for a fuel pin are:

$$M_{UO_2} = \rho_{UO_2} \cdot \pi \frac{d_p^2}{4} H \quad M_U = M_{UO_2} \frac{u_U}{u_{UO_2}} \quad (3.2)$$

where the real density of the fuel is evaluated as $\rho_{UO_2} = \rho_{th} f_{th}$.

For both types of impurity, the moles of gas are computed as:

$$n_g = \frac{c_g \cdot 10^{-6} M_{UO_2}}{u_g} \quad (3.3)$$

with molar masses evaluated as $u_{N_2} = 0.028$ kg/mol and $u_{H_2O} = 0.018$ kg/mol. Regarding the generation of fission gases, it is necessary to evaluate the total number of fissions occurring over the reactor lifetime, and subsequently isolate the fraction responsible for producing Xe and Kr :

$$N_f = \frac{BU \cdot M_U}{E_f} \quad \text{and} \quad n_{Xe+Kr} = N_f \cdot Y \cdot \frac{R_r}{N_A} \quad (3.4)$$

where $N_A = 6.022 \cdot 10^{23}$ 1/mol is the Avogadro number.

Table 3.2: Moles of gas in the plenum.

Species	Variable	Moles [mmol]
Fission gas (Xe + Kr)	n_{Xe+Kr}	62.098
Nitrogen impurity (N ₂)	n_{N_2}	2.091
Water vapor impurity (H ₂ O)	n_{H_2O}	9.756
Total	n_{tot}	73.945

By applying the ideal gas law, the minimum volume required to accommodate all released and initial gaseous species can be evaluated as:

$$V_{min} = \frac{n_{tot} \cdot R \cdot T}{p_{max}} \quad (3.5)$$

where $R = 8.314 \text{ J/(mol K)}$ is the universal gas constant, $T = 346.1 \text{ }^\circ\text{C}$ (equal to 619.25 K) is the temperature evaluated at the upper boundary of the fuel pin, and $p_{max} = 30.85 \text{ MPa}$ is the maximum allowable internal pressure determined via the verification in Section 2.2.

Since the effective expansion volume is restricted to the upper region of the rod, neglecting the clearance gap, the minimum plenum height and the final total height are determined as:

$$H_{min} = \frac{4V_{min}}{\pi D_{ci}^2} \quad \text{and} \quad H_{plenum} = H_{min} + H_{spring} \quad (3.6)$$

where $H_{spring} = 15 \text{ cm}$ is added to account for the structural allowance of the upper plenum holding spring.

Table 3.3: Gas plenum sizing results.

Quantity	Symbol	Value	Unit
Plenum volume	V_{min}	12.34	cm ³
Height (without spring)	H_{min}	22.50	cm
Total height (with spring)	H_{plenum}	37.50	cm

Finally, the ideal gas assumption must be verified for H₂O by comparing the partial pressure evaluated via the ideal gas law:

$$p_{\text{H}_2\text{O},id} = \frac{n_{\text{H}_2\text{O}} \cdot R \cdot T}{V_{min}} \quad (3.7)$$

with the real pressure determined through the IAPWS-95 equation of state (implemented via the **CoolProp** library) under the same molar density and temperature conditions.

Table 3.4: Ideal gas assumption check for water vapor.

Quantity	Variable	Value	Unit
Ideal H ₂ O partial pressure	$p_{\text{H}_2\text{O},id}$	4.071	MPa
Real H ₂ O partial pressure	$p_{\text{H}_2\text{O},real}$	3.775	MPa
Percentage deviation	ε	7.85	%

Treating the water vapor as an ideal gas introduces only a small deviation on its partial pressure (see Table 3.4). As H₂O represents only $\sim 13\%$ of the total moles, the impact on the total plenum pressure is negligible, so the ideal-gas assumption is valid for sizing purposes.

Chapter 4

Conclusions

In this study, a thermo-mechanical safety verification of an AP1000 fuel pin cladding was performed in compliance with the ASME design-by-analysis criteria [3].

The primary membrane stress state, induced by the pressure differential between the internal fission gases and the external coolant ($p_o = p_{sys} = 15.51$ MPa), was evaluated using Lamé’s solutions for thick-walled cylinders. Under nominal operation, the maximum equivalent primary stress intensity is 120.32 MPa, safely below the allowable limit for Zircaloy-4 ($S_m = 137.67$ MPa).

The coupled analysis then accounted for the secondary thermal stresses produced by the steep radial temperature gradient. Even with the superposition of compressive stresses at the inner wall and tensile stresses at the outer wall, the total stress field satisfies the Service Level A limit ($3S_m = 413.0$ MPa).

Lastly, the upper gas plenum was sized to manage the internal pressurization from the released fission products (Xe, Kr) and the initial impurities (N_2 , H_2O), leading to a minimum height of 22.50 cm. Adding the hold-down spring (15.0 cm), a final design height $H_{plenum} = 37.50$ cm was established. The ideal-gas assumption was verified against the IAPWS-95 formulation via **CoolProp**, showing a minor deviation (7.85%) on the H_2O partial pressure.

In conclusion, the analyzed fuel element satisfies all the structural safety requirements imposed by modern Generation III+ nuclear standards.

Text reviewed and corrected with the support of Artificial Intelligence.

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